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Lucari et al.

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(54) **SUCTION-SECURITY SYSTEM**

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(51) **Int. Cl.**

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E05C 19/00	(2006.01)
E05B 35/00	(2006.01)
E05B 13/00	(2006.01)
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E05B 65/52	(2006.01)
E05B 73/00	(2006.01)
E05B 51/02	(2006.01)

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CPC ***E05C 19/00*** (2013.01); ***E05B 13/002*** (2013.01); ***E05B 35/008*** (2013.01); ***E05B 47/0038*** (2013.01); ***E05B 51/02*** (2013.01); ***E05B 65/52*** (2013.01); ***E05B 73/00*** (2013.01); *E05B 47/0045* (2013.01); *F16B 47/006* (2013.01)

(58) **Field of Classification Search**

CPC .. E05B 13/002; E05B 35/008; E05B 47/0038; E05B 65/52; E05B 47/0045; E05C 19/00; F16B 47/006

USPC 248/205.7, 205.8, 309.3, 362, 363

See application file for complete search history.

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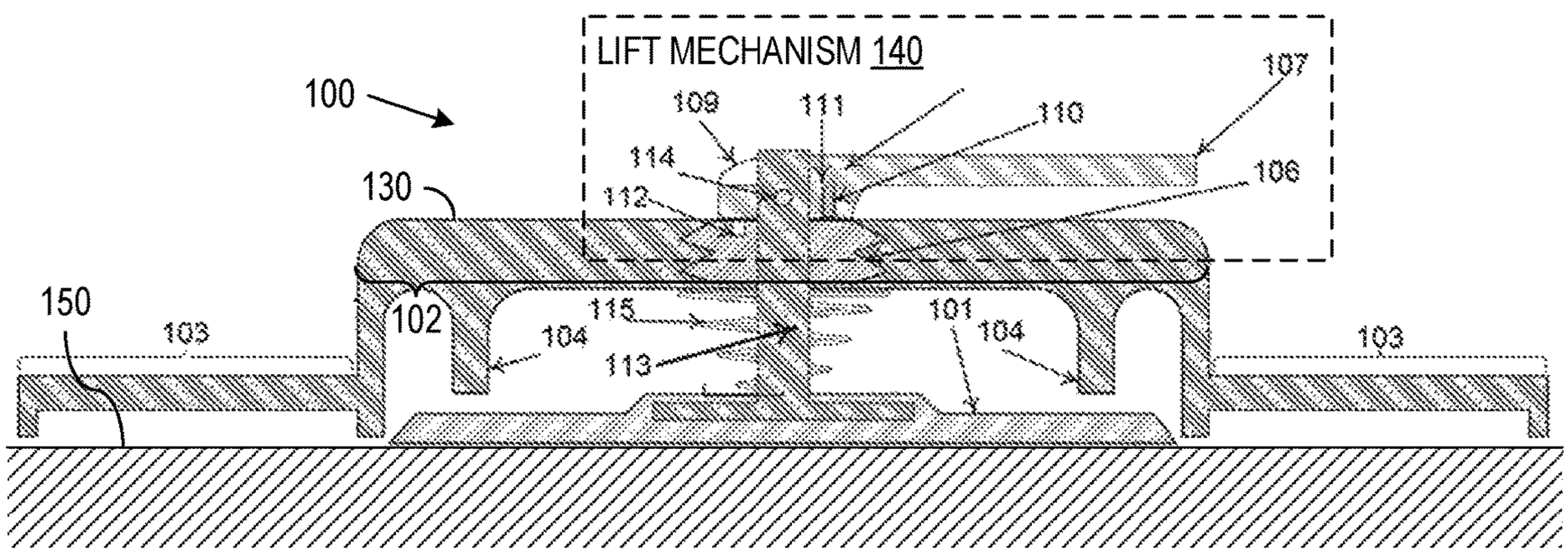
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(57) **ABSTRACT**

A suction-security system includes an adhesion membrane coupled to a swing arm. Rotation of the swing arm lifts a center portion of the adhesion membrane, creating suction and adhesion of the suction-security system to an anchor surface. The swing arm may have a low profile enabling use of the suction security system in an enclosure with minimal loss of available interior space. The system may include a shield protecting the adhesion membrane, and a bushing in the shield permits rotation of a portion of the shield without disturbing adhesion of the adhesion membrane to the anchor surface. The suction-security system may be used in consumer items such as purses, backpacks, briefcases, and suitcases.

17 Claims, 6 Drawing Sheets



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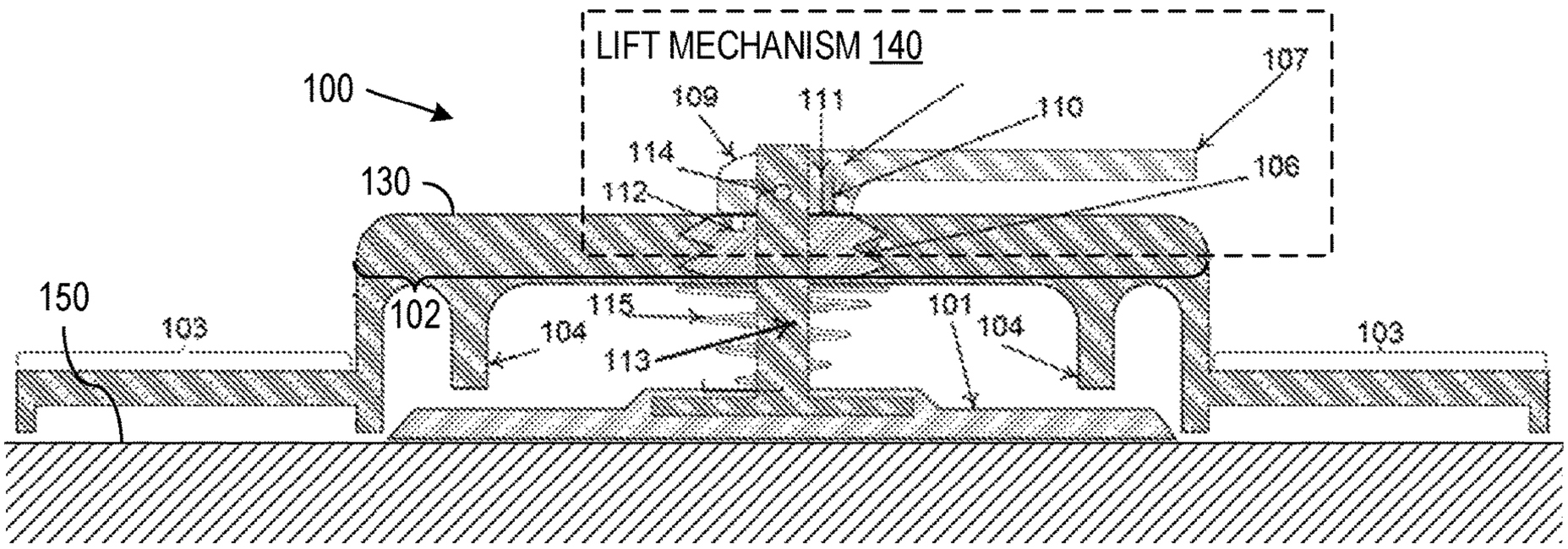


FIG. 1A

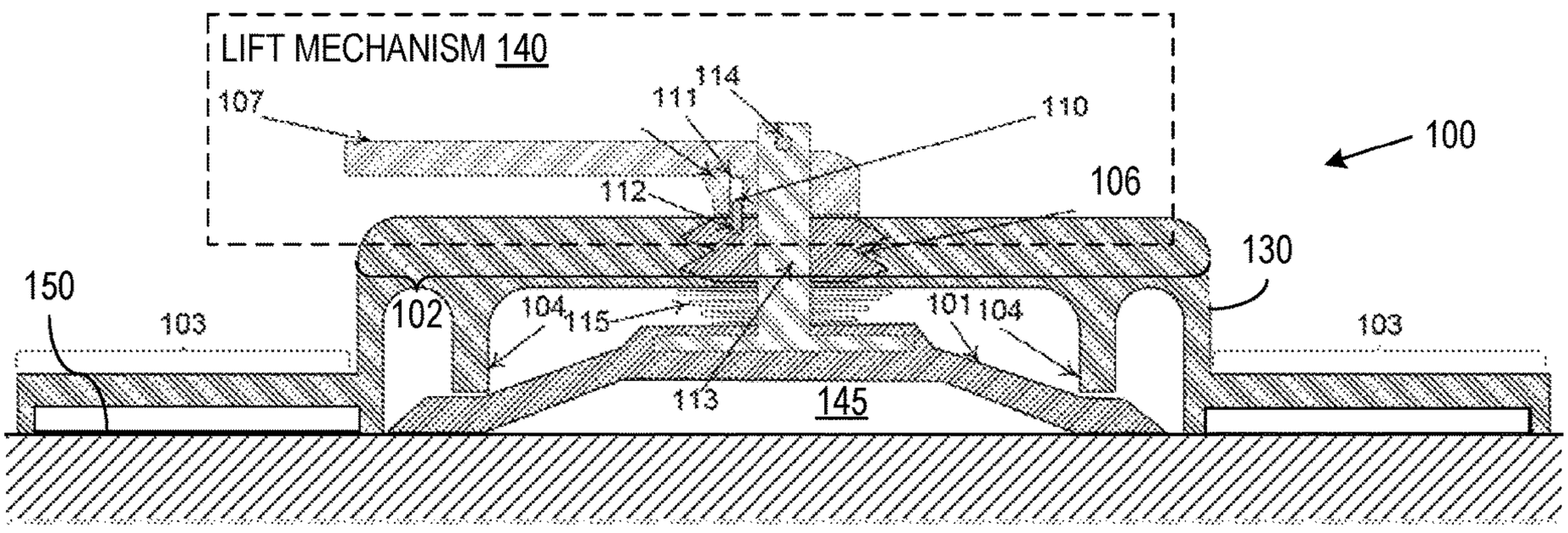


FIG. 1B

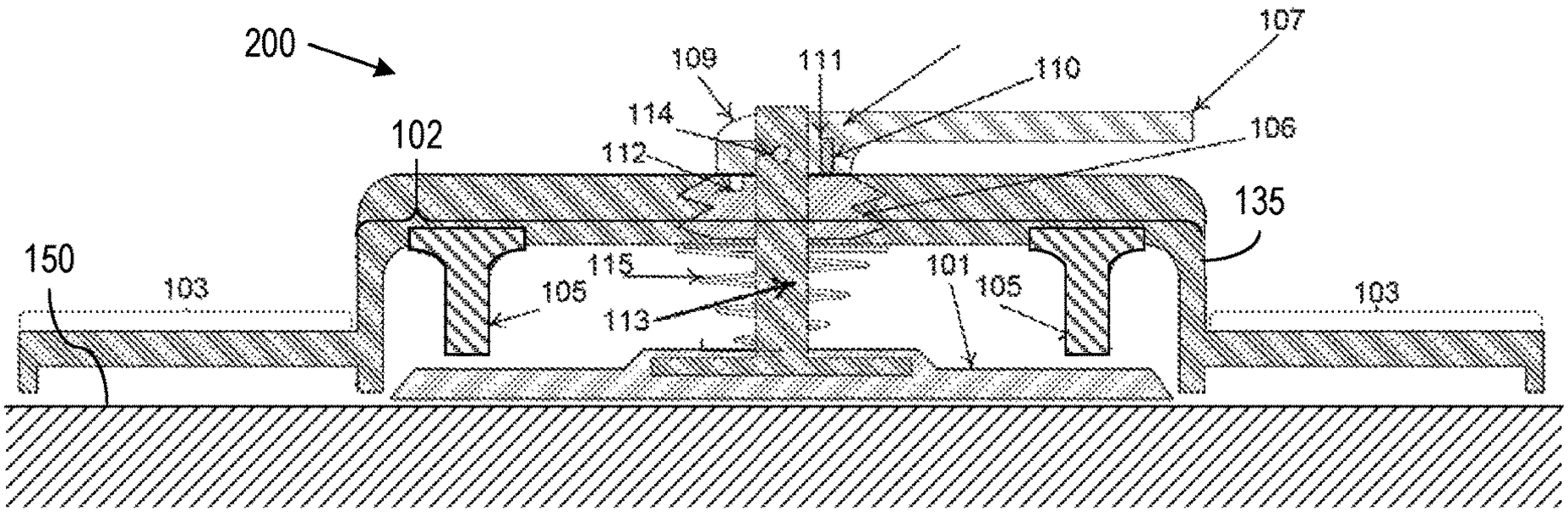


FIG. 2

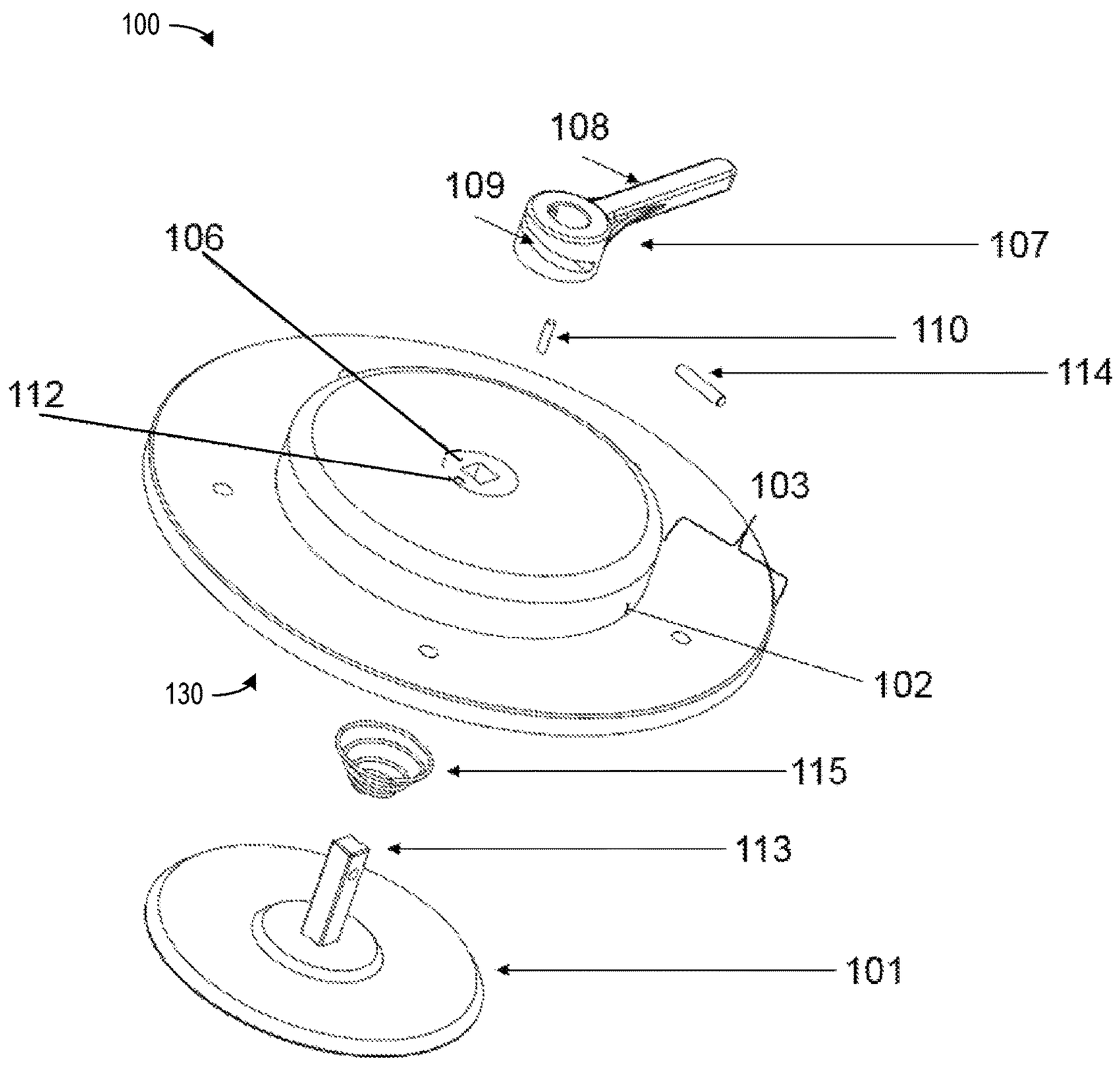


FIG. 1C

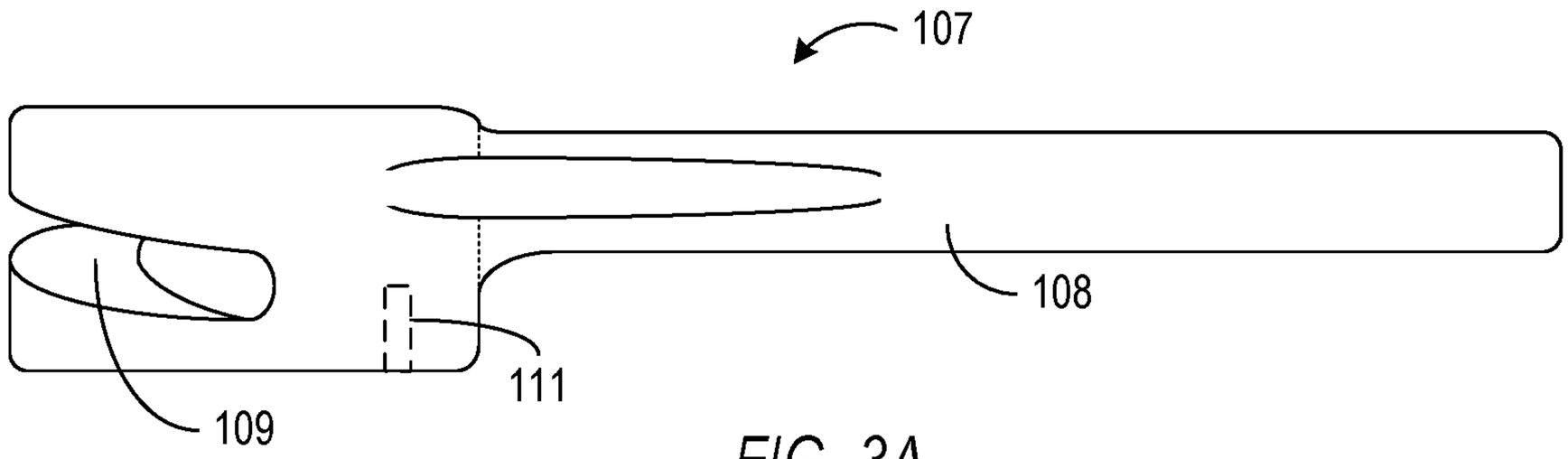


FIG. 3A

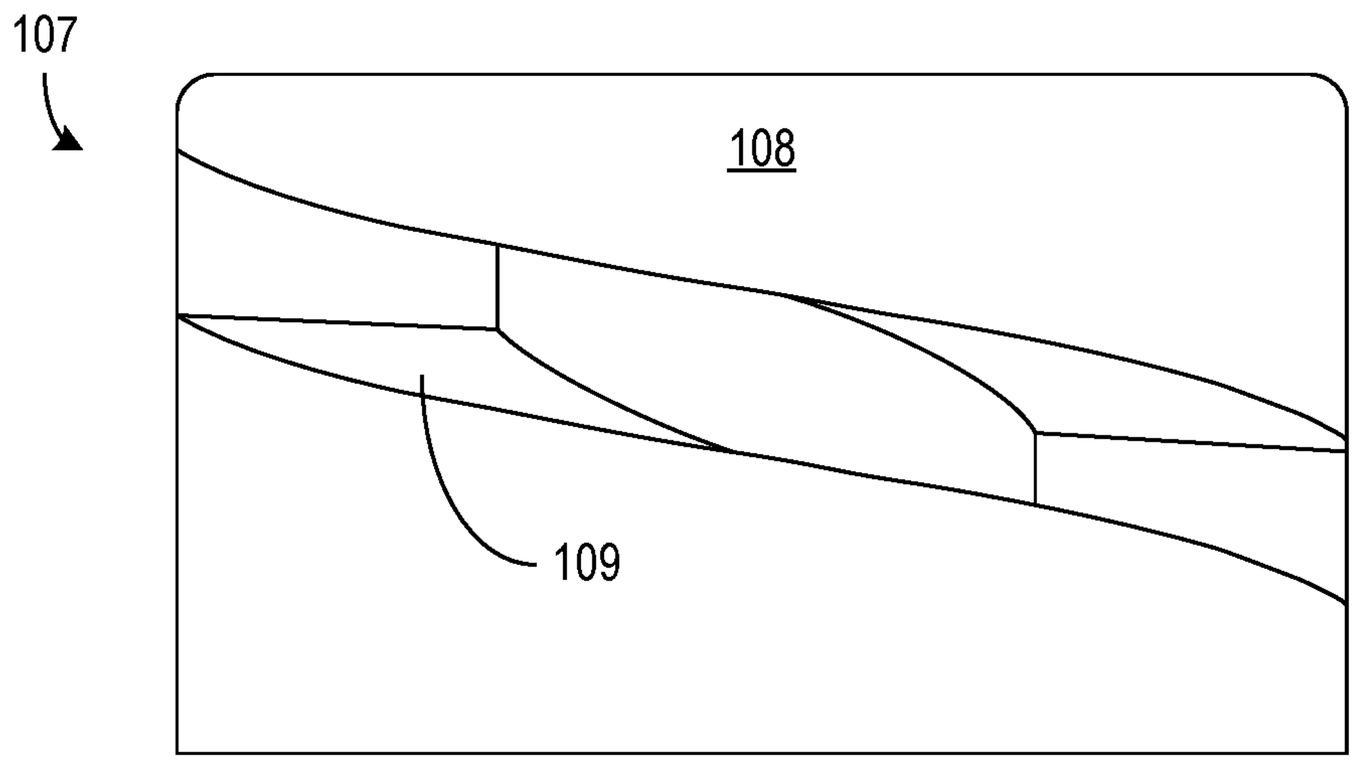


FIG. 3B

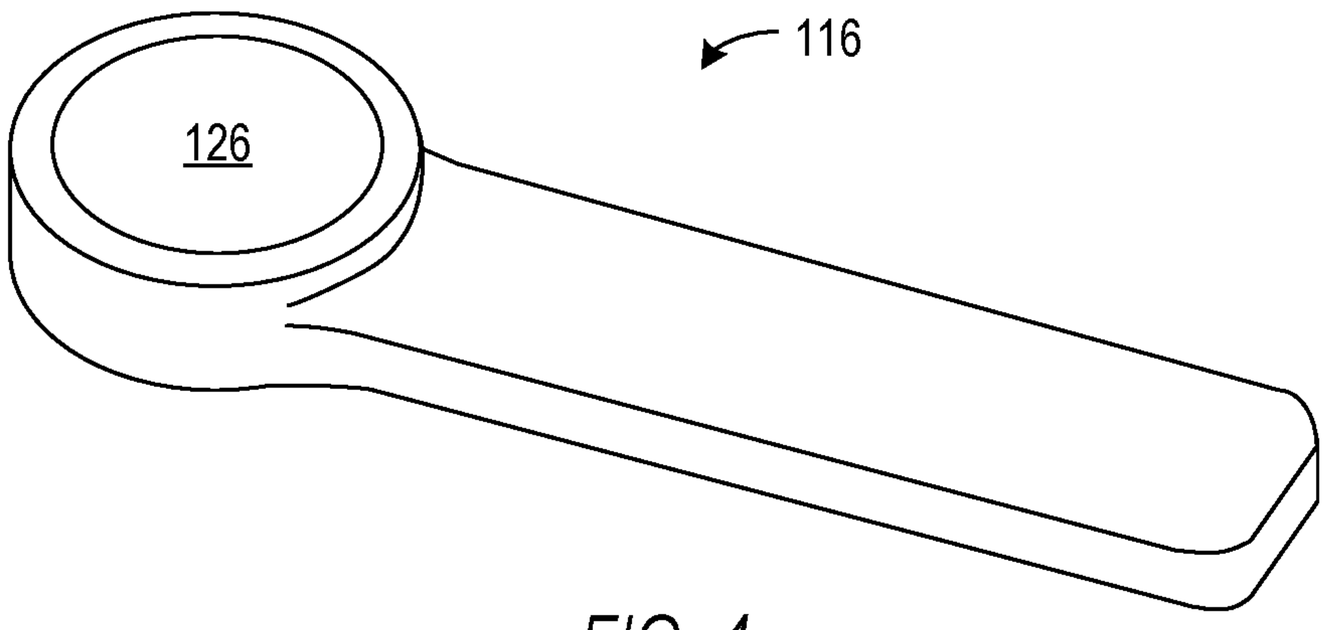


FIG. 4

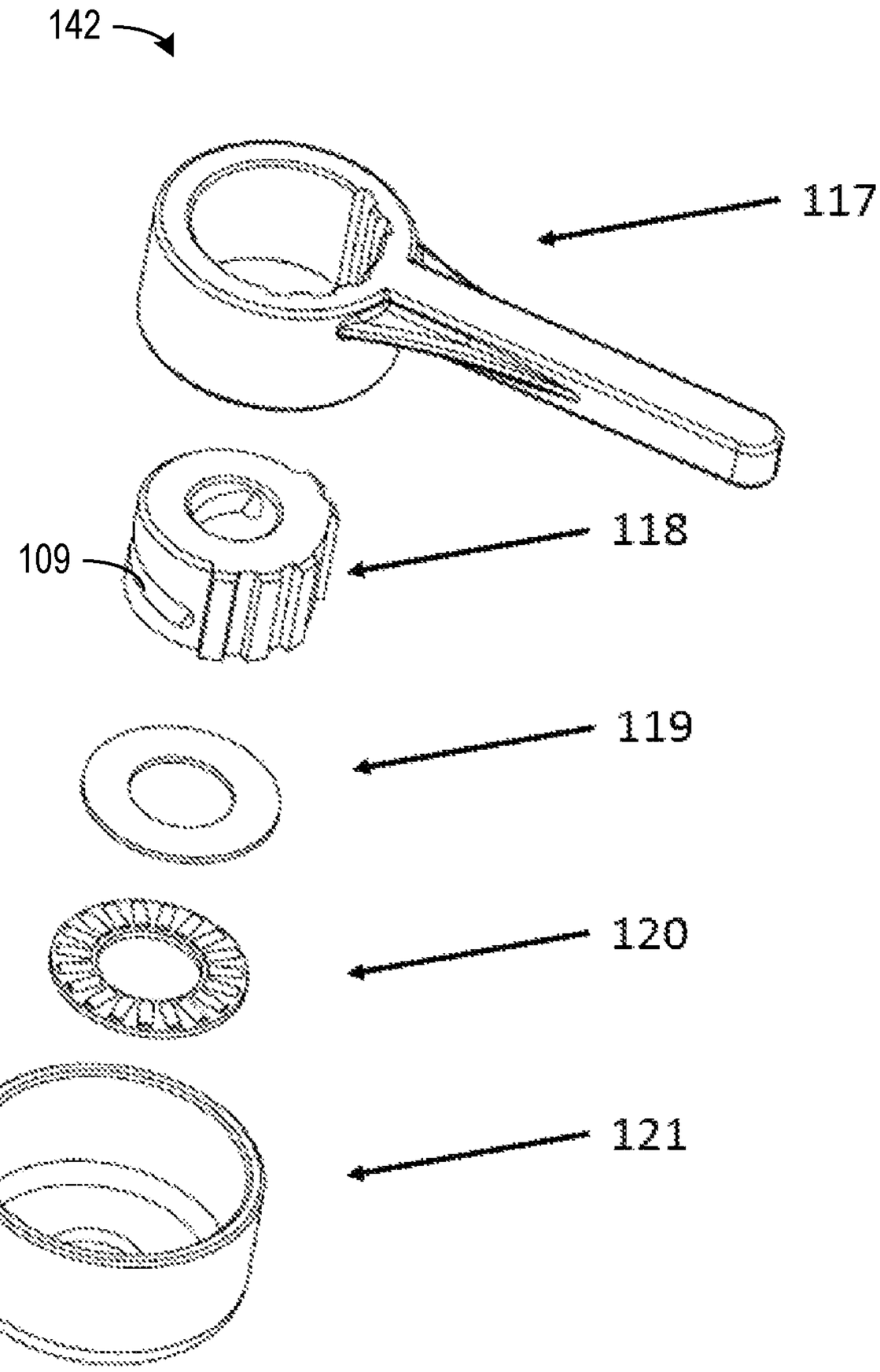


FIG. 5

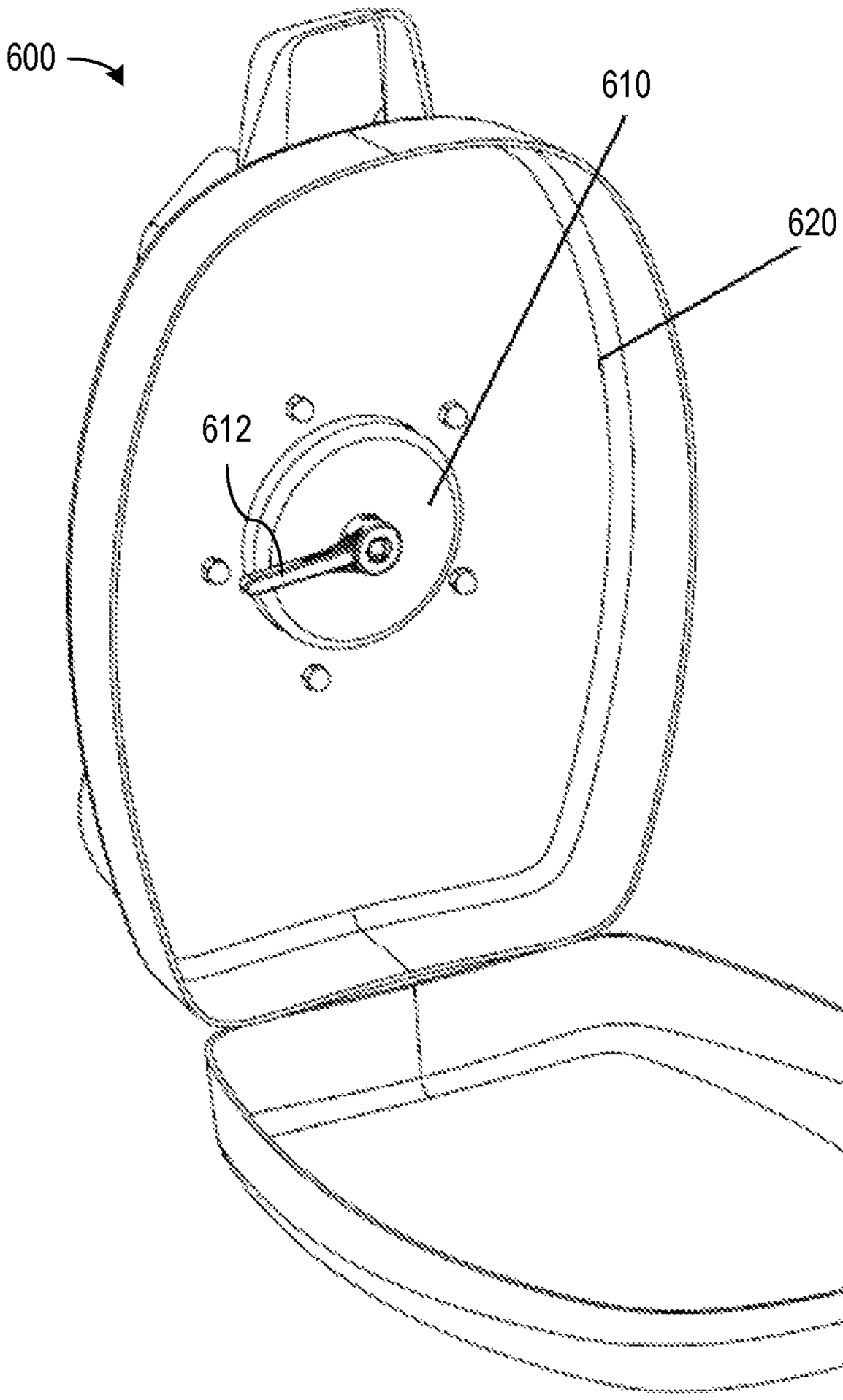


FIG. 6

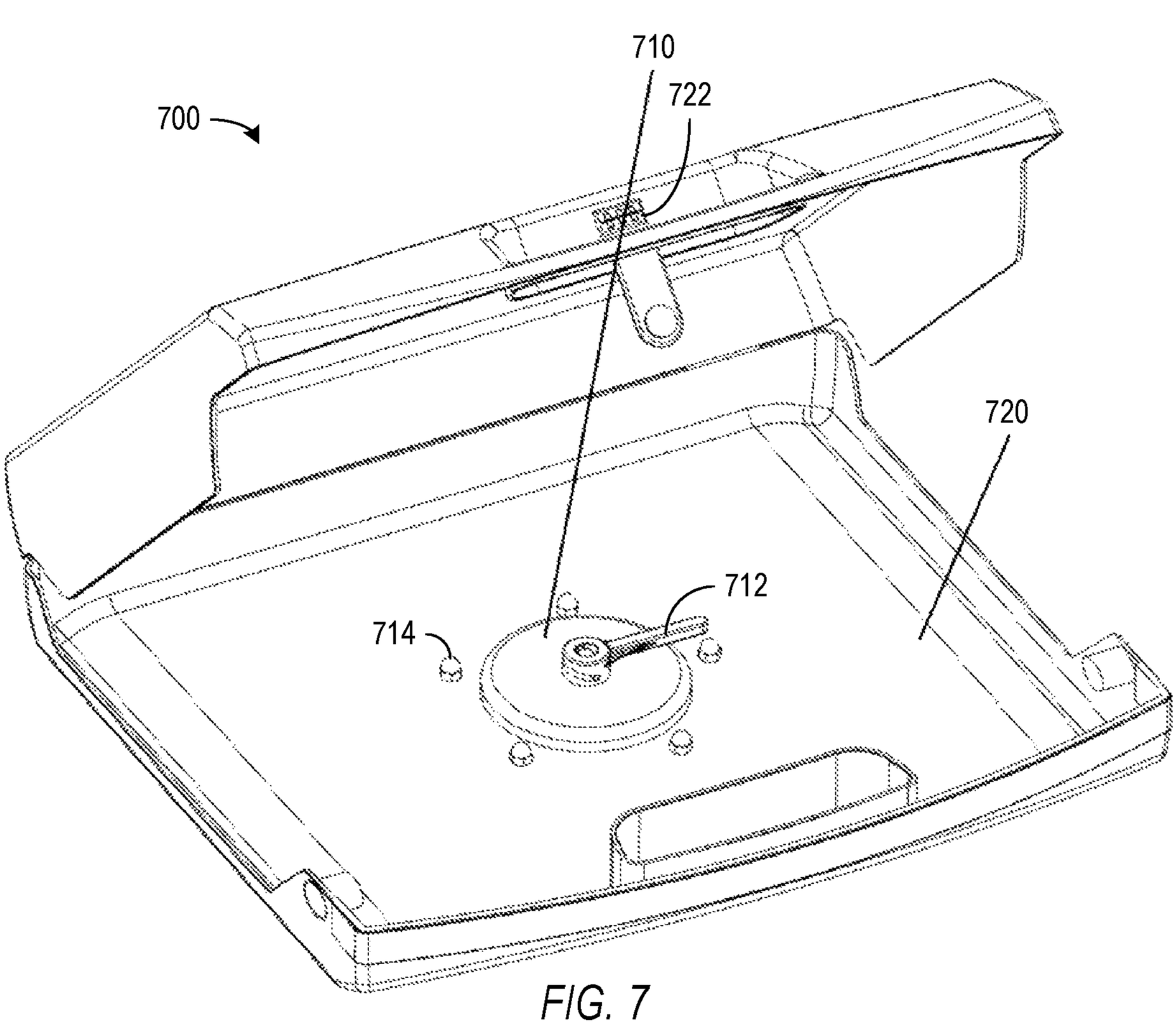


FIG. 7

SUCTION-SECURITY SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

This patent document is a claims benefit of the earlier filing date of U.S. provisional Pat. App. No. 62/564,166, filed Sep. 27, 2017, which is hereby incorporated by reference in its entirety.

BACKGROUND

Some security or antitheft systems secure items by making the items difficult to move. For example, a bicycle may be attached to an anchor such as a tree or a bicycle rack using locks and chains or cables, making the bicycle difficult to move away from the anchor. Similar cables and locks have been used to security electronic devices such as computers. Such security systems generally required specific types of immobile anchors that may be unavailable or inconvenient to use in many locations.

Many techniques have been used to attach items together, but few of the known attachment techniques have been suitable and convenient for portable security or antitheft systems. In particular, a portable security attachment system would ideally be usable in a wide variety of different environments, would be simple to activate, would be secure against unauthorized removal, and yet simple for an authorized user to release.

SUMMARY

In accordance with an aspect of the invention, a suction-security system may be built into or otherwise attached to an item such as a purse, a suitcase, a briefcase, a laptop case, backpack, or other item needing to be secured. The suction-security system may be deployed on any sufficiently smooth surface and activated to create suction that prevents the removal or movement of the item from the surface. The suction-security system may, for example, adhere to any relatively smooth surface such as a table or a tiled or wood floor. Use of the suction-security system may thus be widely available without need for an anchor specifically designed for security and may be available where and when an attachment point for a traditional cable or chain is unavailable or will not work. An activation/deactivation mechanism of the suction-security system may be inside an enclosure defined by the item, and the enclosure, suction-security system, or both may be lockable to prevent unauthorized deactivation. The suction-security system may thus prevent theft.

In accordance with another aspect of the invention, a portable suction-security system can provide a low-profile suction mechanism that may be quickly activated for a secure or stable attachment that an authorized user may quickly release. A suction-security system may particularly include a swing arm coupled to an adhesion membrane via a post, and rotation of the swing arm in a horizontal plane, i.e., about an axis extending through the post, lifts the post and lifts a central portion of the adhesion membrane to create or release suction. The swing arm and the rotations of the swing arm for activation or deactivation are compact and may have minimized height for a low profile mechanism. The swing arm may optionally include a lock mechanism such as a removable key or a magnetically activated locking pin.

In accordance with yet another aspect of the invention, a shield or anti-pry structure surrounding an adhesion membrane in a suction-security system may be attached to the adhesion membrane using a bushing that permits rotation of the shield relative to the adhesion membrane. The bushing connection may thwart attempts to rotate the adhesion membrane to break suction. In particular, rotation of accessible portions of a suction-security system, e.g., the shield, the anti-pry structure, or any external components affixed to shield, do not rotate the adhesion membrane, allowing the adhesion membrane to maintain secure adhesion to an anchor surface. The shield may optionally include a wall ring that presses on a perimeter of the adhesion membrane during activation of suction, and the wall ring may be rotatable relative to external portions of the shield so that the wall ring does not apply torque to the adhesion membrane when the shield is rotated.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1**A is a cross-sectional view of an embodiment of a suction-security system when security suction is disengaged.

FIG. **1**B is a cross-sectional view of the suction-security system of FIG. **1**A when a rotational ramp swing arm is rotated to engage suction.

FIG. **1**C is an exploded view of the suction-security system of FIG. **1**A.

FIG. **2** is a cross-sectional view of an embodiment of a suction-security system with a rotatable depression wall ring.

FIG. **3**A is a side view of a rotational ramp swing arm.

FIG. **3**B is the front view of the rotational ramp swing arm of FIG. **3**A.

FIG. **4** shows a magnetic key fob for use with a suction-security system having magnetically keyed activation mechanism.

FIG. **5** shows an exploded view of an example keyed swing arm for use in a suction-security system.

FIG. **6** shows an example of a backpack including a suction-security system with an activation mechanism inside an enclosure of the backpack.

FIG. **7** shows an example of the case including a suction-security system with an activation mechanism inside an enclosure of the case.

The drawings illustrate examples for the purpose of explanation and are not of the invention itself. Use of the same reference symbols in different figures indicates similar or identical items.

DETAILED DESCRIPTION

A low-profile high-adhesion suction-security system can create a suitable vacuum for adhesion of the suction-security system to a less moveable anchor surface. The suction-security system may have a low profile that is compact and permits use in applications having minimal vertical clearance. One particular embodiment employs a rotational ramp swing arm that aids in providing the low-profile required for use inside thin enclosures such as in a briefcase while still providing lift of a membrane to create or activate high-adhesion suction. The suction-security system is thus ideal for low-profile applications where an enclosure or platform requires attachment to a less moveable anchor surface, but the suction-security systems is also applicable for security where a low profile is not required.

The following discussion is directed to various embodiments. Although one or more of these embodiments may be

discussed in detail, the embodiments disclosed should not be interpreted or otherwise used as a limit to the scope of the disclosure including the claims. One skilled in the art will understand that the following description has broad application, and the discussion of any embodiment is meant only to be an example and is not intended to suggest that the scope of the disclosure, including the claims, is limited to that example.

FIGS. **1**A, **1**B, and **1**C show a suction-security system **100** in accordance with one embodiment of the invention. FIG. **1**A particularly shows a cross-section of suction-security system **100** on an anchor surface **150** before adhesion is activated. FIG. **1**B shows a cross-section of suction-security system **100** after adhesion is activated to adhered suction-security system **100** to anchor surface **150**. FIG. **1**C shows an exploded view illustrating components of suction-security system **100**. Suction-security system **100** particularly includes an adhesion membrane **101**, a lift mechanism **140** attached to adhesion membrane **101**, and a shield or anti-pry structure **130** surrounding membrane **101**.

Adhesion membrane **101** may be a disk or otherwise shaped membrane of a flexible material such as a semi-firm rubber or silicone material with a shape that matches a depression wall **104** that may be part of anti-pry structure **130** surrounding adhesion membrane **101**. In an example implementation, adhesion membrane **101** is circular and about 3 inches in diameter, but adhesion membrane **101** can alternatively be sized and shaped to meet an application's requirements, e.g., having a size and shape matching those of an object being secured or to provide a desired force of adhesive. In general, the material used in adhesion membrane **101** may be selected for flexibility, tackiness, and strength. The size or diameter of membrane **101** may be selected so that the external air pressure on adhesion membrane **101** when suction is activated is sufficient to provide a desired force of adhesion to resist movement of suction-security system **100**. The thickness of membrane **101** may be selected to provide sufficient strength to resist tearing or warping when an expected level of external force is applied to move suction-security system **100**. The thickness of membrane **101** may also be related to the geometry of depression walls **104** that press the perimeter of membrane **101** against anchor surface **150** during engagement of suction-security system **100**.

Shield or anti-pry structure **130** includes a central portion **102** that surrounds the top surface of membrane **101** and may further include an optional extension **103**. Anti-pry structure **130** may be made of a hardened polyvinyl chloride (PVC) or other ridged and sturdy lightweight material. With the illustrated implementation of suction-security system **100**, anti-pry structure **130** may be molded form one-piece structure including central portion **102**, extension **103**, and depression wall **104**. Alternatively, as shown in FIG. **2**, a suction-security system **200** can include an alternative shield or anti-pry structure **135** in which central portion **102** and optional extension **103** are parts of a one-piece structure that is a separate component from a depression wall ring **105**, and depression wall ring **105** may be free to rotate relative to central portion **102** and optional extension **103**. Accordingly, comparing suction security systems **100** and **200**, depression wall ring **105** used in suction-security device **200** of FIG. **2** replaces depression wall **104** of suction-security system **100**. Depression wall ring **105** may ride firmly yet easily within a channel in an external portion of anti-pry structure **135** allowing adhesion membrane wall ring **105** to rotate within the external portion of anti-pry structure **135**, mitigating the chance of suction loss during an attempt to twist or rotate suction-security system **200**.

Various applications of suction-security device **100** or **200** can use or omit optional extension **103**. In some implementations, an enclosure, container, platform, backpack or similar object can attach to suction-security system **100** or **200** particularly by attaching to extension **103**. In other implementations, extension **103** is not required, and an enclosure, container, platform, backpack or similar object can attach directly to the central portion **102** of anti-pry structure **130** or **135**. Holes can be drilled to provide for rivets or bolts to hold an enclosure, container, platform, backpack, or other object to central portion **102** or extension **103** depending upon security requirements. Attachments that are formed within central portion **102** or depression wall **104** or **105** of anti-pry structure **130** or **135** may be sealed to help avoid loss of vacuum under adhesion membrane **101**.

Anchor surface **150** is not part of suction-security system **100** but may be a surface of any convenient relatively immobile object such as a table, a wall, or a floor. More generally, anchor surface **150** may be any surface suitable for sealing against adhesion membrane **101** and allowing creation of the suitable vacuum required for operation. Surface types may be, but are not limited to, glass, tile, vinyl, linoleum, marble, fiberglass, PVC, varnished wood, or other such surfaces that are without irregularities that allow leakages through the seal between adhesion membrane **101** and surface **150**.

FIG. **1**A shows suction-security system **100** before lift mechanism **140** is activated to attach or adhere suction-security system **100** to anchor surface **150**. Lift mechanism **140**, when activated, lifts a central area of membrane **101** as shown in FIG. **1**B, while a perimeter of membrane **101** may be held or pressed against anchor surface **150**, e.g., by depression walls **104** of suction-security system **100** or by depression wall ring **105** for suction-security system **200**. The lifting of the central area of membrane **101** creates suction, e.g., a suitable vacuum, in a volume **145** between membrane **101** and anchor surface **150** so that membrane **101**, suction-security system **100**, and any object attached to suction-security system **100** adheres to anchor surface **150** and is difficult to move.

Lift mechanism **140** in suction-security system **100** employs a membrane lifting post **113**, which may be molded into adhesion membrane **101** or may otherwise be attached to a central area of membrane **101**. Membrane lifting post **113** needs to lift the center of adhesion membrane **101** in a manner that keeps a seal for the vacuum created in volume **145** when the center of adhesion membrane **101** is lifted. Membrane lifting post **113** may be constructed out of a strong material such as a hard aluminum or steel as membrane lifting post **113** may be stressed during a lifting of adhesion membrane **101**.

In the illustrated implementations, membrane lifting post **113** is spring loaded with an adhesion membrane relaxer spring **115**. Adhesion membrane relaxer spring **115** may be a resilient helical spring, and in some applications, spring **115** should be able to support no less than about 40,000 cycles. The number of cycles more generally depends on the application or expected use and useful life of suction-security system **100**. Adhesion membrane lifting post **113** fits and extends through the center of the adhesion membrane relaxer helical spring **115** and through the center hole of a reinforcement bushing **106** mounted in anti-pry structure **130**. Reinforcement bushing **106** may particularly be molded into anti-pry structure **130** and may be constructed of a strong material such as a hard aluminum or steel as

reinforcement bushing **106** may be stressed by the movement of adhesion membrane lifting post **113** and by rotational ramp swing arm **107**, particularly where rotational ramp swing arm **107** meets the top of adhesion membrane lifting post **113**. Whether or not reinforcement bushing **106** is molded into anti-pry structure **130**, reinforcement bushing **106** may, in some implementations, be knurled or may have other or additional features to hold bushing **106** firmly to anti-pry structure **130**.

Reinforcement bushing **106** may have a channel or hole drilled and used as a security locking pin reinforcement bushing channel **112**. The diameter of channel **112** may be only slightly larger than that of a security locking pin **110** to allow for non-binding movement of locking pin **110** into or out of channel **112**, and channel **112** may be deep enough for full inclusion of pin **110** within channel **112**. Additionally, in some implementations, reinforcement bushing **106** may be made of a non-ferrous material, for example, if an implementation requires security locking pin **110** to be magnetized as described below.

Membrane lifting post **113**, which extends through reinforcement bushing **106**, connects to an activation mechanism that lifts membrane lifting post **113**. In the embodiment of suction-security system **100**, the activation mechanism includes a rotational ramp swing arm **107**, which is rotatable about an axis of membrane lifting post **113**. FIGS. **3**A and **3**B show side and front views of one implementation of rotational ramp swing arm **107** and particularly show a helically inclined ramp surface **109** that may be formed in a hub of rotational ramp swing arm **107**. Implementations using optional security locking pin **110** may have security locking pin swing arm channel **111** provided in rotational ramp swing arm **107** during manufacturing of rotational ramp swing arm **107**. Locking pin swing arm channel **111** may have a diameter and depth only slightly larger than the diameter and length of locking pin **110** to allow for non-binding movement of locking pin **110** into channel **111**, and full inclusion of locking pin **110** in channel **111** when locking pin **110** is not engaged with reinforcement bushing channel **112**.

A portion of membrane lifting post **113** extends into or through the hub of rotational ramp swing arm **107**. A pin **114** in the portion of membrane lifting post **113** extending into rotational ramp swing arm **104** slides on helically inclined ramp surface **109**, so that rotation of rotational ramp swing arm **107** from the position shown in FIG. **1**A to the position shown in FIG. **1**B lifts pin **114**, membrane lifting post **113**, and the central portion of membrane **101**.

Suction-security system **100** may be mechanically to an item defining an enclosure so that lift mechanism **140** is inside the enclosure and membrane **101** and at least a portion, e.g., extension **103**, of anti-pry structure **130** is outside the enclosure. The bottom surfaces of membrane **101** and anti-pry structure **130** can then be placed on anchor surface **150**. Rotation of rotational ramp swing arm **107** causes swing arm lifting ramp surface **109** to lift pin **114** and therefore lifting post **113** vertically, i.e., along the axis of rotation of swing arm **107**. The rotation necessary to activate lift mechanism **140** may be less than a full turn, e.g., about 180° of rotation, which allows for quick and easy application to adhere suction-security device **100** to anchor surface **150**. As an alternative to using pin **114** and helical ramp surface **109**, lifting post **113** and swing arm **107** may respectively have coarse male and female threading that similarly lifts lifting post **113** when swing arm **107** rotates, but pin **114** and ramp surface **109** may provide a stronger or more durable mechanism for rapid lift within a 180° rotation. The rotation of ramp swing arm **107** is in a plane parallel to anchor surface **150** and allows a much lower profile for suction-security system **100** than would a lever mechanism activated by movement in the lift direction. Rotational ramp swing arm **107** may also be implemented as a one-piece structure, which may reduce manufacturing costs when compared to a more complex lift structure, e.g., using multiple screw type mechanisms or lifts requiring high vertical clearance.

Rotational ramp swing arm **107** is stressed primarily at the fulcrum of handle **108**, and on the surface of swing arm lifting ramp surface **109** as ramp surface **109** lifts lifting post pin **114** and pin **114** rides up ramp surface **109** during lifting of adhesion membrane **101**. To more evenly accommodate this load, swing arm lifting ramp surface **109** may be manufactured so that the ramp profile maintains contact with the entire engaged length of the adhesion membrane lifting post ramp pin **114**. Rotational ramp swing arm **107** and adhesion membrane lifting post ramp pin **114** may be designed to be resilient over many use, e.g., no less than about 40,000 cycles; a number that is only suggested and depends on the overall use model of the suction-security system. In some specific implementations, rotational ramp swing arm **107** may be made of stainless steel, hardened aluminum, or hardened polymer, depending upon desired life expectancy, weight, cost, and other factors depended on the intended use of suction-security device **100**. A non-ferrous material would be used in an implementation that employs a locking pin **110** and related components.

Ramp pin **114** may be fit in a hole through adhesion membrane lifting post **113**. In implementations employing a locking pin **110**, the hole through adhesion membrane lifting post **113** for ramp pin **114** may be above the security locking pin channel **112** to allow the lowest part of the swing arm lifting ramp surface **109** to fit under lifting post ramp pin **114**. Lifting post ramp pin **114** may extend the width of the top of adhesion membrane lifting post **113** plus the width of the swing arm lifting ramp surface **109**, so that one end of lifting post ramp pin **114** is flush with the exterior of swing arm lifting ramp surface **109** and the other end of lifting post ramp pin **114** is flush with the exterior of the adhesion membrane lifting post **113**. Lifting post ramp pin **114** should be a suitably strong and inflexible material to accommodate stress during the lifting process along the swing arm lifting ramp surface **109** and at the junction of adhesion membrane lifting post **113**.

A user when beginning to activate lift mechanism **140** to create suction and attachment of suction-security system **100** to anchor surface **150** may apply downward pressure at the top of anti-pry structure **130**, e.g., apply pressure near or on rotational ramp swing arm **107**, so that anti-pry external structure **130** contacts anchor surface **150** and adhesion membrane depression wall **104** of anti-pry structure **130** contacts the top of adhesion membrane **101** around the perimeter of adhesion membrane **101**. Membrane relaxer spring **115** may be under partial tension at this point. Rotation of rotational ramp swing arm **107** by about 180° provides lift of adhesion membrane lifting post **113** through the external structure of reinforcement bushing **106** as swing arm lifting ramp surface **109** forces adhesion membrane lifting post ramp pin **114** vertically upward. A suitable vacuum thus created between adhesion membrane **101** and anchor surface **150** securely attaches suction-security system **100** to anchor surface **150**, and contact between the perimeter of membrane **101** and a sufficiently smooth anchor surface **150** will maintain adequate vacuum and secure attachment for an extended time, e.g., up to a few days or more.

Anti-pry structure **130** may prevent unauthorized release of the vacuum by shielding adhesion membrane **101**, e.g., preventing undesired lifting of the outer edge of adhesion membrane **101**. In general, a base portion of anti-pry structure **130** may contact (or be in close proximity) with anchor surface **150** to prevent insertion of a pry bar or tool under anti-pry structure **130**. Further, anti-pry structure **130** may include extension **103**, extending well beyond the edges of central portion **102** surrounding membrane **101**. In one specific implementation, adhesion membrane **101** and the central portion of anti-pry structure **130** may be about 4 inches in diameter, and optional extension **103** may extend about 2 inches beyond the perimeter of membrane **101**. Extension **103** may be shorter, e.g., about ½ inch tall, than central portion **102**, e.g., about ¾ inch tall, since extension **103** does not need to be tall enough to contain adhesion membrane **101**. Extension **103** may further inhibit access to membrane **101** and provide large moment arm to resist prying at the edges of anti-pry structure **130**. The size of the optional anti-pry structure extension **103** may depend on the application in which suction-security system **100** is employed. For example, suction-security system **100** may be employed in an enclosure such as a briefcase that is large enough that the extent of the briefcase sufficiently inhibits access to membrane **101**, in which case anti-pry extension **103** may not be required.

Anti-pry structure **130** may also help prevent an attempt to break adhesion through attempted lateral or rotational movement of suction-security system **100**. In particular, reinforcement bushing **106** may permit externally accessible portions of anti-pry structure **130** to rotate about lifting post **113** and adhesion membrane **101**. Further, some implementations such as suction-security system **200** of FIG. **2** replace membrane depression wall **104**, which may contact adhesion membrane **101**, with rotatable depression wall ring **105**. Depression wall ring **105** may be a ring that resides inside of the remainder of anti-pry external structure **135** of suction-security system **200**. As a result, if an unauthorized person manages to rotate an external portion of anti-pry structure **135**, wall ring **105** and membrane lifting post **113** do not rotate with the external portion of anti-pry structure **135** and do not transfer the rotation to adhesion membrane **101**. Suction-security system **200** can remain attached to anchor surface **150** even when rotated. Suction-security system **200** of FIG. **2** may otherwise be identical to suction-security system **100** of FIGS. **1**A, **1**B, and **1**C as described herein.

Suction-security system **100** may further employ a security mechanism to prevent unauthorized rotation of swing arm **107**. The illustrated embodiment of suction-security system **100** employs locking pin **110** that fits in swing arm channel **111** in swing arm **107** and is able to at least partially slip into bushing channel **112** in reinforcement bushing **106** when rotation ramp swing arm **107** is in the locked position shown in FIG. **1**B. (The locked position corresponds to the adhered position.) Swing arm channel **111** in rotational ramp swing arm **107**, locking pin **110**, and bushing channel **112** in bushing **106** may be reinforced or made of strong material to resist sheering if rotation of arm **107** is attempted while locking pin **110** bridges both swing arm channel **111** and bushing channel **112**. In one implementation, locking pin **110** is iron or another ferromagnetic material and may be magnetically lifted out of bushing channel **112** using a magnet to unlock rotation. In another implementation, locking pin **110** is magnetized and may be magnetically lifted out of or pushed into bushing channel **112** using a magnet to lock or unlock rotation of rotational ramp swing arm **107**.

FIG. **4** shows an optional magnetic key fob **116** that may be used to move locking pin **110** between swing arm channel **111** and bushing channel **112**. In some implementations, optional magnetic key fob **116** is used to both engage and disengage the rotational ramp swing arm optional security locking pin **110**. Magnetic key fob **116** may be formed of a polymer or other suitable material with a Neodymium or Rare Earth magnet **126** imbedded in fob **116**. Depending on the application, locking pin **110** may also be magnetized to allow magnetic key fob **116** to be turned so that the side of the magnet in fob **116** facing the hub of swing arm **101** attracts locking pin **110** or be turned so that the opposite side of the magnet in fob **116** faces the hub of swing arm **101** and repels locking pin **110**. With a magnetized locking pin **110**, magnetic key fob **116** may be used to push locking pin **110** to bridge swing arm channel **111** and bushing channel **112** to lock rotational swing arm **107** when suction has been activated or may be used to lift (or push) locking pin **110** free of one of swing arm channel **111** and bushing channel **112** to unlock rotational arm **107** and permit rotation to release the suction.

Another optional modification to a rotational ramp swing arm is shown in FIG. **5**. FIG. **5** particularly shows an exploded view of one implementation of a mechanically keyed rotational ramp swing arm **142**. Mechanically keyed rotational ramp swing arm **142** includes a removable rotational ramp swing arm keyed socket **117** that a user can separate from the remainder of keyed rotational ramp swing arm **142** to thereby prevent the rotation of swing arm lifting ramp surface **109**. In FIG. **5**, swing arm lifting ramp surface **109** is within a swing arm lifting ramp keyed nut **118**, which is turned by coupling rotational ramp swing arm keyed socket **117** to swing arm lifting ramp keyed nut **118** and rotating rotational ramp swing arm keyed socket **117**. Swing arm lifting ramp keyed nut **118** rides on a thrust washer **119** and a needle bearing **120** through which membrane lift post **113** passes. A rotational ramp swing arm key guard **121** surrounds swing arm lifting ramp keyed nut **118** to prevent or inhibit use of a tool, other than rotational ramp swing arm keyed socket **117**, from engaging with and rotating swing arm lifting ramp keyed nut **118**. FIG. **5** only shows one example of a key pattern, but a sufficient number of alternative patterns may be provided in different suction-security systems to reduce duplications of the same key patter potentially being utilized in close proximity to each other. Mechanically keyed rotational ramp swing arm **142** can be used in place of rotational ramp swing arm **107** in suction-security system **100** or **200**.

Rotational ramp swing arm keyed socket **117** and swing arm lifting ramp keyed nut **118** may be made of a stainless steel, hardened aluminum, or hardened polymer, depending upon desired life expectancy, weight, cost, and other factors. The thrust washer **119** is a common load bearing element and can be made of the same materials. Thrust needle bearing **120** is typically a hardened chrome steel and should be able to handle a minimum of 40,000 cycles; a number that is only suggested and should be validated within the overall use model of the suction-security system. The rotational ramp swing arm key guard **121** may be stainless steel so as to resist breaking with common hand tools. In general, the rotational ramp swing arm optional security locking pin **110**, swing arm channel **111**, and bushing channel **112** are not required or used in implementations using mechanically keyed rotational ramp swing arm **142**.

Suction-security systems such as disclosed herein may be employed to secure any desired object, e.g., through attachment cables or other structures that connect a suction-

security device to the object being secured. In some implementations, a suction-security system may be built into the object being secured. FIG. **6** shows an example of a low-profile suction-security system integrated into a backpack **600**. In particular, suction-security system **610** may be mounted through and attached to a wall of an enclosure **620** of backpack **600**. Suction-security system **610** may be in accordance with any of the implementations disclosed herein and is mounted so that an activation control **612** of a lift mechanism of suction security system **610** is located in the interior of enclosure **620**. Activation portion **612** may particularly include a rotational ramp swing arm as described above. An adhesion membrane (not visible in FIG. **6**) of suction-security system **610** is located outside of enclosure **620**. Backpack **600** may be secured by pressing the external adhesion membrane and external portions of an anti-pry structure against a suitable anchor surface and rotating activation control **612** to activate suction. For a backpack **600** in which enclosure **620** may not be lockable, suction-security system **610** may be magnetically lockable or mechanically lockable as described above to prevent unauthorized removal or movement of backpack **600**.

Suction security systems may similarly be applied in other types of flexible material enclosures such as a purse. In such uses, whether the suction-security system is lockable or not, the suction security system may still prevent "snatch and go" thefts of the purses, backpacks, and other flexible material enclosures by preventing movement of the enclosure from an anchor surface.

FIG. **7** shows a case **700** incorporating a low-profile suction-security system **710** mounted in a wall of an enclosure **720** of case **700**. Case **700** may be a hard-walled structure such as a briefcase, a laptop case, or a suitcase. Suction-security system **710** may be a low-profile suction-security system of minimized height so that the remaining space in enclosure **720** is sufficient to secure desired contents of case **720**. Suction-security system **710** may be in accordance with any of the implementations disclosed herein and is mounted so that an activation control **712** of a lift mechanism of suction-security system **710** is located in the interior of enclosure **720**. An adhesion membrane (not visible in FIG. **7**) of suction-security system **710** is located outside of enclosure **720**. Case **700** may be secured by pressing the external adhesion membrane and external portions of an anti-pry structure of suction-security system **710** against a suitable anchor surface, rotating activation control **712** to activate suction, and then closing enclosure **720** so that activation control **712** is enclosed within briefcase **700**. Briefcase **700** further has a lock **722** that may be locked to prevent unauthorized access to activation control **712**. Accordingly, suction-security system **710** does not need to be lockable but may optionally be magnetically lockable or mechanically lockable as described above to prevent unauthorized removal or movement of briefcase **700**.

In one implementation, suction-security system **710** is identical to suction-security system **100**, and central portion **102** of anti-pry structure **130** suction-security system **100** extends through a hole in a wall of enclosure **720**. Fasteners **714**, e.g., bolts or rivets, also extend through respective holes in the wall of enclosure **720** and fasten to extension **103** of anti-pry structure **130** of suction-security system **100**. With a low profiled system, the portion of suction-security system **100** inside enclosure **720** may be about 1.5 inches tall or less.

In another implementation, a clear or windowed case with an integrated suction security system can be employed in a store such as a jewelry store or other commercial environment that sells items that may need to be secured while being displayed. For example, case **700** may have a transparent or windowed top, so that enclosure **720** can provide a secure semi-portable display case for expensive items. Case **700** may be moved and secured as needed for potential buyers to observe items in case **700**, for example, during a side-by-side comparisons with items in another case. The securable mobility of case **700** may be more convenient than having items separated and secured at great distance from each other inside of large glass cases.

The suction-security systems as disclosed herein may be used to secure items against theft as described above and may also be used to secure items for use. In particular, an item may be secured to an anchor surface in a moving environment to prevent shifting of the item. In an automobile or a plane, for example, the anchor surface may be a tray table or the back of a seat, or in a boat on open and rough seas, the anchor surface may be the deck of the boat. The suction-security systems may also be used to provide a stable working platform in working environments, for example, where technicians may hold equipment in place by attaching a suction system of the equipment to smooth surfaces such as the surfaces of junction boxes or panels. For such attachments, the anchor surface may be horizontal, inclined, or even horizontal, and the suction may still fix the item in place.

Although suction-security systems have been explained in specific uses, there are many other use cases, modifications, and variations possible further implementations within the scope of the following claims.

What is claimed is:

1. A suction-security system comprising:
an adhesion membrane;
a shield surrounding a top of the adhesion membrane;
a post coupled to a central portion of the adhesion membrane, the post extending through the shield;
a pin set in the post; and
a swing arm including a hub through which a portion of the post extends, the hub including a helical inclined surface on which the pin rides such that rotation of the swing arm by less than a full turn about an axis extending through the post lifts the post in a direction along the axis and lifts the central portion of the adhesion membrane sufficiently to adhere the adhesion membrane to an anchor surface in contact with a portion of the adhesion membrane.

2. The system of claim **1**, wherein the swing arm comprises:
a keyed nut including the helical inclined surface; and
a socket shaped to detachably engage the keyed nut during the rotation of the swing arm.

3. The system of claim **1**, further comprising a locking pin, wherein:
when the swing arm is in a position lifting the central portion of the adhesion membrane, the locking pin is movable between a first channel in the swing arm and a second channel in the shield; and
the locking pin extending simultaneously into both the first channel and the second channel prevents rotation of swing arm.

4. The system of claim **3**, wherein the locking pin is ferromagnetic and movable out of the second channel using a magnet above the swing arm.

5. The system of claim **3**, wherein the locking pin is magnetized and responds to a magnet above the swing arm

by moving in a first direction into the first channel or moving in a second direction into the second channel depending on an orientation of the magnet.

6. The system of claim **1**, wherein the shield comprises a wall positioned to contact a top surface of the adhesion membrane adjacent to a perimeter of the adhesion membrane.

7. The system of claim **6**, wherein the wall is sized and positioned to press on the perimeter of the adhesion membrane when the shield and membrane are pressed against a flat surface.

8. The system of claim **6**, wherein the wall comprises a ring positioned to contact the top surface of the adhesion membrane adjacent to the perimeter of the adhesion membrane, the ring being rotatable relative to other portions of the shield.

9. The system of claim **1**, wherein the shield comprises a bushing through which the post extends, the bushing making at least a portion of the shield rotatable about the post.

10. The system of claim **1**, wherein the shield comprises:

a central portion surrounding the top of the adhesion membrane; and

an extension extending from the central portion.

11. The system of claim **10**, wherein the extension comprises mounting structure for attachment of the suction security system to an enclosure with the extension being outside the enclosure and the central portion extending into an interior of the enclosure.

12. The system of claim **1**, further comprising an enclosure, wherein the shield is attached to the enclosure and positioned so that the adhesion membrane is outside the enclosure and the swing arm is inside the enclosure.

13. The system of claim **12**, wherein the enclosure is selected from a display case, a briefcase, a laptop case, a suitcase, or a backpack.

14. The system of claim **12**, wherein when adhered to the anchor surface, the suction system provides the enclosure with stability from shifting.

15. A suction-security system comprising:

an adhesion membrane;

a shield surrounding a top of the adhesion membrane;

a post coupled to a central portion of the adhesion membrane, the post extending through the shield;

a swing arm coupled to the post such that rotation of the swing arm about an axis extending through the post lifts the post in a direction along the axis and lifts the central portion of the adhesion membrane in the direction along the axis; and

a locking pin, wherein:

when the swing arm is in a position lifting the central portion of the adhesion membrane, the locking pin is movable between a first channel in the swing arm and a second channel in the shield; and

the locking pin extending simultaneously into both the first channel and the second channel prevents rotation of swing arm.

16. The system of claim **15**, wherein the locking pin is movable out of the second channel using a magnet adjacent to the swing arm.

17. The system of claim **1**, wherein engagement of the pin with the inclined surface limits rotation of the swing arm to less than a full turn about the axis extending through the post.

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